

Office of Biomedical Advanced Research and Deveopment Authority
Division of Research, Innovation & Ventures (DRIVe)
Easy Broad Agency Announcement (EZ-BAA) BAA-22-100-SOL-00003



The purpose of Amendment #019 is the following:

1) Reopen the following Area of Interest (AOI):

AOI #15: ReDIRECT

2) Revise the following Area of Interest:

AOI #15: ReDIRECT

INTRODUCTION AND OVERVIEW INFORMATION

A. Development Opportunity Objective:

Under this Amendment, DRIVe is doing the following:

- 1) Reopening the following research Area of Interest (AOI):

AOI #15: ReDIRECT

- 2) Revising the following research Area of Interest (AOI):

AOI #15: ReDIRECT

We are seeking abstract submissions for the following AOI:

AOI #15: ReDIRECT (Repurposing Drugs In Response to Chemical Threats)

The availability of effective medical countermeasures (MCMs) against chemical threats is critical in the treatment of their acute health effects. Necessary attributes of effective MCMs against chemical threats include ease of administration during a mass-casualty situation and rapid efficacy as a post-exposure therapy. Drug repurposing is a strategy that is used to identify new uses, outside of their original clinical indication, for Food and Drug Administration (FDA) approved or late-stage investigational therapeutics. The identification of existing compounds for repurposing as MCMs holds the potential to expand current response capabilities to chemical threats, as well as potentially mitigating the costs and risks associated with conventional drug discovery.

Biomedical Advance Research Development Authority (BARDA) is requesting abstract submissions for projects that repurpose existing therapeutics as MCMs against chemical threats (cyanide, opioids, nerve agents, chlorine, sulfur mustard, etc.). These therapeutics should have a strong mechanistic justification for potential use as MCMs. Ideal candidates should have a known safety profile from previous clinical indications or development and be safe and effective for the entire population, including at-risk populations such as pediatrics, geriatrics, pregnant women, and immunocompromised individuals. MCM candidates should:

- 1) Be FDA-approved or in advanced clinical development for a conventional indication like the pathophysiology associated with exposure to a chemical agent; and
- 2) Utilize delivery routes or mechanisms that improve ease of administration (including, but not limited to, reformulation of existing products) to large numbers of exposed individuals during mass casualty situations. Priority will be given to products manufactured in the United States.

Therapeutics that are eligible for drug repurposing may target any of the following:

Pulmonary Agents: Development of MCMs to prevent and treat lung damage (including pulmonary edema, acute respiratory distress syndrome, pneumonitis, and fibrosis) resulting from exposure to agents such as chlorine, sulfur mustard and phosgene.

Opioids: Development of MCMs to treat life-threatening respiratory depression caused by opioid overdose. These post-exposure treatments should be quick-acting and effective against a variety of opioids, including synthetic opioids such as Fentanyl. Candidates must have a mechanism of action other than opioid receptor antagonism. Priority consideration will be given to candidates that are also effective against respiratory depression caused by non-opioid pharmaceutical-based agents.

Vesicants: Development of MCMs that limit harmful aspects of exposure to vesicating agents such as sulfur mustard and Lewisite. Preference is given to drugs with potential to ameliorate the long-term effects of exposure including Mustard Gas Keratopathy.

Knockdown Agents/Cellular Asphyxiants: Development of MCMs to treat acute poisoning from agents such as cyanide, sulfides, azides, etc. which disrupt tissue oxygen utilization. Antidotes should be easily administered by first responders in personal protective equipment. Priority consideration will be given to treatments that are also effective against smoke inhalation-related exposure.

Nerve Agents and Organophosphorus (OP) Pesticides: Repurposing of MCMs to treat life-threatening effects and long-term sequelae of nerve agents and OP pesticides. Antidotes should be easily administered by first responders in personal protective equipment.

Computational approaches to identify candidates for drug repurposing: Development of improved methods to rapidly identify FDA approved or late-stage candidate compounds that can be repurposed against any of the aforementioned chemical threats.

To be considered responsive under this AOI, respondents should have:

- 1) A drug that is a candidate for repurposing as a MCM against pulmonary agents, opioids, vesicants, cellular asphyxiants nerve agents, or organophosphate pesticides;
- 2) A clear commercial indication for the drug that is separate from its development as an MCM
- 3) **Plans to market the therapeutic for a primary indication separate from its use as a medical countermeasure.** Projects that propose therapeutics that will *only* be used as MCMs (even for more than one threat) are considered nonresponsive to this AOI.
- 4) At a minimum, an open Investigational New Drug Application and in-process Phase 1 clinical trial. Please note that preference will be given to companies that have an FDA approved drug, or one that has completed Phase 1 or 2 trials as evidenced by a clinical study report: and
- 5) A clear rationale as to why the candidate should be efficacious as a post-exposure chemical MCM.

All potential respondents are *highly encouraged* to reach out to the program team for a market research call prior to submission.

B. Eligible Respondents & Scope Parameters:

This Amendment is open to all responsible sources as described in the EZ-BAA. Abstract submissions that do not conform to the requirements outlined in the EZ-BAA may be considered non-responsive and will not be reviewed. An entity must have an active

registration with <https://sam.gov> at the time of submission to be reviewed. If not, the abstract submission will not be reviewed and will be rejected. Please do not attempt to submit an abstract if your registration is not active in <https://sam.gov>.

IMPORTANT NOTE: Interested vendors are strongly encouraged to request and schedule a pre-submission call before submitting an abstract. This request should include the project title, key project staff, and a brief description of the proposed project. Please submit the requests to the following:

AOI #15: ReDIRECT (chemrepo@hhs.gov)

The closing date for abstract submissions for this AOI is listed below.

Area of Interest	Closing Date for Abstract Submissions
#15	12:00pm ET on May 3, 2024

Note: To streamline the EZ-BAA, all Areas of Interest will be open for a few months at a time following a staggered approach. This is being done to encourage high-quality submissions earlier in the fiscal year allowing adequate review time. Depending on programmatic need and funding availability, Areas of Interest may be reopened for another period of time.

C. Number of Awards:

Multiple awards are anticipated and are dependent upon the program priorities, scientific/technical merit of abstract submissions, how well the abstract submissions fit within the goals of the AOI, and the availability of funding. The program funding is subject to change based on the Government's discretion.

Funding is limited, so we encourage any interested vendors to reach out to the respective program as soon as possible before submitting an abstract.

D. Amendment Application Process:

This Amendment will follow the same submission process and review procedures as those established under this EZ-BAA, unless otherwise noted. For complete details, please read the EZ-BAA in its entirety along with all amendments.

IMPORTANT NOTE: Respondents who are awarded a contract under each of these AOIs will be required to share any collected, de-identified data to advance the field and knowledge. Interested Respondents are strongly encouraged to commercialize their technology and algorithms, however, note that consistent with BARDA's mission and federal standards, data collected through the use of government funding will be delivered to BARDA for government usage pursuant to applicable regulations and law.